JONAH SCHEPERS

Data Scientist | M.S. Data Science, Virginia Tech (4+1) | Available Summer 2027

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Builds AI evaluation systems: an LLM-as-judge pipeline now moving into production and four threat classifiers scored on PR-AUC across 525,000 connections. Python, SQL and applied statistics, with error-weighted validation and uncertainty bounds behind every result. Stays with a problem until the answer holds up on a second dataset, then presents it to the people who act on it.

EXPERIENCE

Graduate Assistant | Analytics & AI, Virginia Tech *Aug 2026 to Present*

- Moving the evaluation framework from prototype to production behind a **Click CLI**, with **Pydantic** strict-schema config validation, per-model rate limiting and error isolation so analysts outside the project can run evaluations unassisted
- Gating every prompt and configuration change on a golden-dataset regression run before it ships, backed by a **pytest** suite and Ruff linting
- Onboarding new institutional-research use cases onto the config-package pattern, encoding **time-censored cohort handling**, **small-sample suppression** below 10 students, and privacy guardrails that decline individual-student lookups against aggregate data

Data Science Intern | Analytics & AI, Virginia Tech *May 2026 to Aug 2026*

- Built a configuration-driven **LLM-as-judge evaluation pipeline** in Python on the **Strands Agents SDK** and **Amazon Bedrock** that benchmarks AI agents across a matrix of models, prompts and rubrics, then emits per-model JSON scores and an HTML comparison report
- Moved domain tools, fixture data, test cases and scoring rubrics into versioned config packages, making the pipeline **use-case agnostic** so a new evaluation ships with zero code changes
- Curated a **48-case golden dataset** pairing each question with a verified answer and the tool call it should trigger, then wrote weighted rubrics with **hard gates** that zero any response fabricating a statistic or quoting a rate for a milestone the cohort has not reached
- Instrumented runs with **OpenTelemetry** trajectory capture over deterministic fixture-backed tools, so the judge model audits every tool call and parameter and every run reproduces exactly
- Benchmarked **six models** across Amazon Bedrock and VT's ARC-hosted LLM API, then extended the pipeline to **multi-turn grading** that tests whether an assistant holds its privacy and data-censoring rules under five turns of user pushback

Research Assistant | Department of Physics, Virginia Tech *Nov 2023 to Dec 2025*

- Built a **Python ETL pipeline** over raw telescope FITS imagery to test a star-formation model against galaxy NGC 2764, deprojecting every pixel to a galactocentric radius and merging five image maps into one **385-point** modeling table that carries per-pixel measurement error
- Fit a **multi-Gaussian expansion** to Spitzer and 2MASS surface photometry to build the stellar-density predictor, then re-ran the full pipeline on both surveys to confirm the fit was not an artifact of a single instrument
- Wrapped the physics model in a custom **scikit-learn estimator** so **Grid Search** could tune the clumping factor against a chi-square objective weighted by per-observation measurement error, moving it from the model default of 5.0 to 2.65
- Validated with error-aware **7-fold Cross-Validation**, propagating observational error through **100 Monte Carlo draws per fold**: **12.8% RMSE**, reduced chi-square of 1.70, and residuals split 46/54 over- and under-prediction with no systematic bias
- Quantified parameter uncertainty with **100-iteration bootstrap** refits and bounded worst-case model error at **17.7%** under pessimistic scenario analysis

AI Trainer, Data Annotator | Handshake AI *Nov 2025 to Present*

- Built CAD models as training data for benchmarking AI spatial reasoning and object recognition
- Wrote prompts and rubrics that grade AI-generated CAD reconstruction and video synthesis on 3D-reasoning quality and output fidelity
- Annotated image and video datasets used to train AI object-detection models

SELECTED PROJECTS

Ozmo Agentic AI Call Center Pipeline *2026*

Python | RAG | Local LLM | BERTopic | UMAP | LLM-as-Judge

- Built a Python ETL pipeline that pulls and cleans call center conversation data for NLP analysis
- Built a RAG layer on a local LLM that replicates Ozmo's production agentic AI for offline testing
- Clustered conversations with **BERTopic** over **UMAP** dimensionality reduction, then scored sentiment and benchmarked agent performance with an LLM-as-judge framework

Network Intrusion Detection & Deployment Business Case *2026*

Python | XGBoost | Class Imbalance | ROC / PR-AUC | Cost-Benefit Modeling

- Trained four **XGBoost** classifiers to separate advanced persistent threats, phishing, zero-day exploits and insider activity from normal traffic across a **525,000-connection** evaluation set where every attack class sat below **0.4% prevalence**

- Scored the models on **precision-recall AUC**, ROC curves and confusion matrices, since 0.4% prevalence makes accuracy meaningless: **0.83 PR-AUC** on insider threat, the hardest of the four classes, at a **0.3% false-positive rate** against a 2% business requirement
- Built a deployment simulator and cost-benefit model that turns detection gains into dollars, then presented the case to a stakeholder panel: **$1.2M** build cost, break-even inside year one, **3-year NPV of $3.15M** at an 8% discount rate

Manufacturing Demand Forecasting & Optimization *2025*

PostgreSQL | ARIMA | Linear Programming | CRISP-DM

- Extracted and transformed manufacturing data with PostgreSQL, then built an **ARIMA** model to forecast boom-lift demand
- Designed a **linear program** that minimizes overproduction and aligns output with the demand forecast
- Ran the project on the CRISP-DM lifecycle, from business understanding to a deployment-ready model

Insurance Subrogation Prediction (NLP) *2025*

Python | NLP | Random Forest | Regex

- Built a **random forest classifier** that predicts which vehicle insurance claims need subrogation, at **86% accuracy**
- Engineered NLP features from unstructured claim notes using stop-word removal, stemming, and lemmatization
- Wrote regex extraction that pulls liability percentages out of free-text claim narratives

EDUCATION

Virginia Tech, Blacksburg VA | M.S. Data Science (4+1 Program) *Expected May 2027*

Virginia Tech | B.S., Computational Modeling and Data Analytics *2023 to Fall 2026*

AWARDS & COMPETITIONS

2nd Place | CMDA Data Competition *2024*

- Built a model that flags fraudulent vehicle insurance claims in Python and R, beating dozens of teams at **over 95% accuracy**
- Ranked fraud indicators by feature importance and presented the findings to a panel of faculty judges

5th Place | ASA DataFest Competition *2024*

- Analyzed statistics-course data under NDA in Python at the national competition, identifying learning patterns tied to student performance
- Recommended which chapters most affected grade outcomes

TECHNICAL SKILLS

Languages: Python (4 yrs), R (3 yrs), SQL, C, Java, MATLAB, CUDA, Bash, LaTeX
ML & Data Science: ML, NLP, LLM-as-Judge Evaluation, Model Benchmarking, LLM Fine-Tuning, Prompt Engineering, Agentic AI, RAG, Bayesian/Regression, ARIMA, BERTopic, UMAP, CRISP-DM
Tools & Math: Git, Linux, AWS Bedrock, Strands Agents SDK, MCP, Pydantic, Databricks, MPI, OpenMP, Linear Algebra, Differential Equations, Advanced Statistics